Optimal Trading Filters: a Wiener-Hopf Approach

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Abstract

We give a closed-form solution to the problem of maximizing expected gain from a return-predicting signal, net of the impact cost of one's own trades, under a propagator model with a general symmetric positive-definite kernel. The key tool is a projected-inverse identity: for any causal-anticausal factorization $C = C_- C_+$ of the cost operator, the adapted inverse satisfies

$$(P_+ \, C \, P_+)^{-1} \; = \; C_+^{-1} \, P_+ \, C_-^{-1}$$

on the adapted subspace L^2_{adap}. This identity holds in the Wiener–Hopf setting on the whole line and in the Gohberg–Krein/Arveson outer-factorization setting on a finite interval, and reduces the adapted optimal trading rate to a three-step composition: anticausal whitening of the signal's forecast curve, adapted projection, and causal coloring. Existing treatments characterize the adapted optimum implicitly, through stochastic Volterra equations of the second kind; the present approach yields an explicit operator formula whose form depends on the kernel only through its factorization. For the empirically supported power-law kernel $G(t) = |t|^{-\beta}$, $\beta \in (0, 1)$, the factors are Liouville fractional integrals of order $\nu = (1 - \beta)/2$, and the optimal rate reduces to a composition of anticausal and causal Marchaud fractional derivatives of order ν applied to the signal's forecast curve. On a finite horizon, the same recipe produces a boundary-deformed fractional derivative that converges to the whole-line formula in the interior at rate $O(d(t)^{-\nu})$, where $d(t)$ is the distance to the nearest boundary.

1 Introduction

1.1 The gain–cost problem

Institutional trading of large orders — index rebalances, algorithmic strategies against short-horizon forecasts, hedges against derivative books — routinely moves quantities that are large multiples of top-of-book depth. Empirical studies of trade-and-quote data (Lillo et al. 2003, Bouchaud et al. 2004, Gatheral 2010, Jusselin and Rosenbaum 2020) establish that each child order pushes the mid-price against the trader's direction and that the resulting *transient impact* is well described by a translation-invariant propagator kernel $G(t-s)$ that decays with lag: the price at time t receives a contribution $\int_{s \le t} G(t-s) \, u_s \, ds$ from the trader's own past rate u. The cost of a large parent order therefore depends on the whole path of its execution schedule, not only on its instantaneous rate.

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When the trader holds a return-predicting signal α_t — a forecast of price change conditional on time-t information — the gain from trading in the signal's direction competes with the impact cost of moving into the position. Under the propagator model of Bouchaud, Gefen, Potters and Wyart (2004), extended by Gatheral (2010), the expected P&L of an adapted trading rate $u \in L^2_{\mathrm{adap}}$ is a linear gain against a quadratic cost, and the *gain–cost problem* is

$$\max_{u \in L^2_{\mathrm{adap}}} \; \mathbb{E}\int u_t\, \alpha_t\, dt \;-\; \tfrac{\gamma}{2}\, \mathbb{E}\iint G(|t-v|)\, u_t\, u_v\, dt\, dv, \tag{1}$$

with $\gamma > 0$ a cost-aversion coupling. Write C for the symmetric convolution $(Cu)(t) = \int G(|t-v|)u_v\, dv$, positive-definite ($\hat{C}(\xi) \geq 0$) to rule out static round-trip arbitrage. Two kernels run through the paper: the empirically supported power law $G(t) = |t|^{-\beta}$, $\beta \in (0,1)$ estimated at 0.2–0.6 across markets and asset classes (Lillo et al. 2003, Bouchaud et al. 2004, Gatheral 2010, Jusselin and Rosenbaum 2020), with symbol $\hat{C}(\xi) = c_\beta |\xi|^{\beta-1}$ and $c_\beta = 2\Gamma(1-\beta)\sin(\pi\beta/2)$; and the tractable exponential $G(t) = e^{-\kappa|t|}$, $\kappa > 0$, with symbol $2\kappa/(\kappa^2+\xi^2)$. The power-law symbol has a low-frequency singularity and C is unbounded on $L^2(\mathbb{R})$; §4 gives the technical domain fix required for the pure power-law setting. The exponential kernel and the finite-horizon truncation to $[0,T]$ are bounded on L^2 without further regularization.

1.2 Analogy with Markowitz portfolio choice

Problem (1) has the same convex-quadratic structure as the mean–variance portfolio problem (Markowitz 1952, Merton 1972),

$$\max_{w \in \mathbb{R}^N} \; w^\top \mu \;-\; \tfrac{1}{2}\lambda\, w^\top \Sigma w, \tag{2}$$

under the correspondence

$$(w,\, \mu,\, \Sigma,\, \lambda) \;\longleftrightarrow\; (u,\, \alpha,\, C,\, \gamma).$$

The Markowitz optimum is $w^\star = \lambda^{-1}\Sigma^{-1}\mu$ with value $\frac{1}{2\lambda}\mu^\top\Sigma^{-1}\mu$ (the Mahalanobis norm of expected return, squared). The direct analog $u^\star = \gamma^{-1}C^{-1}\alpha$ would be the optimum of the trading problem before imposing adaptedness and accounting for the non-locality of the Hessian. Two features of the temporal problem block the direct analog:

(i) Non-locality of the Hessian. C is a temporal convolution against a symmetric kernel, so $(Cu)(t)$ depends on u_v for v both before and after t.

(ii) Adaptedness of the feasible set. The trading rate u_t at time t must be $\mathcal{F}_t$-measurable; computing $(C^{-1}\alpha)_t$ requires future values of α which the trader does not have.

1.3 Prior treatments

Existing treatments of problem (1) establish existence, uniqueness, and structural characterizations of $u^\star$.

Deterministic constant-signal case. Almgren and Chriss (2001) treat quadratic cost with the exponential kernel and derive closed-form schedules by state augmentation. Obizhaeva and Wang (2013) solve the same problem in a limit-order-book resilience framework. For the power-law kernel, Gatheral, Schied and Slynko (2012) give the optimal deterministic schedule as an eigenfunction of a Fredholm integral equation, exhibiting the Söhngen–Tricomi endpoint weights $(t(T-t))^{(\beta-1)/2}$. In these papers the "signal" is a Lagrange multiplier fixed by the terminal-inventory constraint; the trader has no additional information.

Stochastic-signal case, exponential kernel. Lehalle and Neuman (2019) solve the Almgren–Chriss problem with a general adapted signal and linear temporary impact, obtaining $u^\star$ as a conditional expectation of the future signal — the temporary-impact cost operator is diagonal, so the projected inverse reduces to the projection itself. Neuman and Voß (2022) extend the analysis to the exponential resilience kernel, using the Markov property to reduce the problem to a finite-dimensional linear-quadratic control with Riccati feedback — available via state augmentation specific to the exponential kernel and not to the empirically supported power-law case. Cartea, Jaimungal and Penalva (2015) develop the broader algorithmic-trading framework.

Stochastic-signal case, general kernel. For a general propagator kernel with a stochastic signal, Abi Jaber and Neuman (2025) prove existence and uniqueness of $u^\star$ and characterize it as the unique solution of a linear stochastic Volterra equation of the second kind: a fixed-point equation involving the kernel and the signal, from which closed forms are recovered only after kernel-specific reductions. The framework extends to matrix-valued cross-impact (Abi Jaber et al. 2024b) and constrained trading including battery-storage applications (Abi Jaber et al. 2024a), inheriting the same Volterra-equation form.

Stochastic-signal case, power-law kernel. Forde, Sánchez-Betancourt and Smith (2022) treat Gaussian signals on $[0, T]$ using the finite-interval factorization $G_T = T T^*$ with T a Volterra operator of Riemann–Liouville type (Porter and Stirling 1990, Ex. 6.2, 9.2); the optimal rate is the integral of the signal against the resolvent of T, inverted via the Chakrabarti–George Abel formula (Chakrabarti and George 1994). The representation is horizon-tied and gives no bulk formula in terms of the signal path.

Deterministic factorization theory. Wiener and Hopf (1931) introduced the factorization of scalar Fourier symbols on a half-line as the basis for solving convolution integral equations. Krein (1962) developed the corresponding half-line integral-equation theory, and Wiener (1949) applied the factorization to stationary linear prediction. Gohberg and Krein (1970) established the finite-interval factorization $I + K = (I + L^*)(I + L)$ for positive kernels via Volterra triangularization, and Arveson (1975) generalized the picture to positive operators on Hilbert spaces equipped with a continuous nest of projections.

The approach of Abi Jaber and Neuman (2025) is general in both the kernel and the signal: existence and uniqueness of the adapted optimum are established for any Volterra propagator and any adapted square-integrable signal, and the optimum is characterized as the unique solution of a linear stochastic Volterra equation of the second kind. This characterization is implicit: closed-form expressions require kernel-specific reductions, and the fractional-calculus structure of the power-law case is not made explicit. The present paper gives a complementary analysis. For the stationary whole-line problem, the Wiener–Hopf approach yields a closed-form operator identity for $(P_+CP_+)^{-1}$: the adapted inverse is expressed directly as the product $C_+^{-1}P_+C_-^{-1}$ with $C = C_-C_+$ the causal-anticausal factorization of the cost operator, without resolving a free-boundary equation. The two approaches differ qualitatively. The Volterra-equation approach handles the full finite-horizon problem with terminal inventory constraints and non-stationary signals; the spectral factorization approach gives an explicit formula for the operator structure of the adapted inverse, from which the fractional-derivative form in the power-law case follows by direct substitution.

Across these strands the adapted optimal rate under a stochastic signal and a power-law kernel is characterized implicitly, through fixed-point Volterra equations or horizon-tied resolvent representations. No bulk closed form in terms of the signal path has been given, and the fractional-calculus structure of the problem is left buried in the resolvent. The factorization identity below closes both gaps.

1.4 Contribution

The paper contributes three things: an abstract identity (Lemma 1), a closed-form solution for the stationary problem (Theorem 1 and Corollaries 1–2), and an explicit formula for the signal-speed response function (§5.1) that settles when contrarian trading is optimal.

Core abstract identity (Lemma 1). Let C be a symmetric strictly positive operator on a filtered Hilbert space $\mathcal{H} = L^2(\Omega \times \mathbb{R})$, and let $C = C_- C_+$ be any factorization into a causal Volterra factor C_+ (kernel on $\{s \leq t\}$) and its adjoint $C_- = C_+^*$. Let P_+ be the orthogonal projection onto the adapted subspace $L^2_{\text{adap}} \subset \mathcal{H}$. The projected-inverse identity

$$(P_+ \, C \, P_+)^{-1} \; = \; C_+^{-1} \, P_+ \, C_-^{-1} \tag{3}$$

holds on L^2_{adap}. The identity is structural: it follows from the two triangularity relations $P_+^{\perp} C_+ P_+ = 0$ and $P_+ C_- P_+^{\perp} = 0$, which are equivalent to causality of C_+, and from strict positivity of C. Its proof (§3.3) is independent of the specific kernel; it covers the Wiener–Hopf factorization on the whole line and the Gohberg–Krein/Arveson outer factorization on a finite interval as special cases. The identity is the temporal analog of the formula $\Sigma^{-1} = L^{-\top} L^{-1}$ for a matrix inverse via its Cholesky factor L, with the adapted projection playing the role of the lower-triangular structure.

Closed-form optimal trading rate (Theorem 1). The adapted first-order condition of (1) is $\gamma \, (P_+ C P_+) \, u^\star = \alpha$. Substituting (3) gives the closed form

$$u^\star \; = \; \gamma^{-1} \, C_+^{-1} \, P_+ \, C_-^{-1} \, \alpha. \tag{4}$$

The acausal operator C_-^{-1} applied to the adapted signal α at time s requires the future of the signal; the adapted projection P_+ replaces this future by the conditional expectation, i.e., the *forecast curve*

$$\bar{\alpha}(t, s) = \begin{cases} \alpha_s, & s \leq t, \\ \mathbb{E}_t[\alpha_s], & s > t. \end{cases}$$

The result (4) then factors as three successive operations on the forecast curve: anticausal whitening (C_-^{-1} acting on $\bar{\alpha}(s, \cdot)$ at s), adapted projection (enforcing $\mathcal{F}_s$-measurability), and causal coloring (C_+^{-1}). This three-step architecture is the temporal analog of the Wiener–Kolmogorov prediction filter (Wiener 1949).

Power-law kernel: bulk formula (Corollary 1). For $G(t) = |t|^{-\beta}$ the whole-line Wiener–Hopf factors are the Liouville (Weyl) fractional integrals $C_\pm = c_\beta^{1/2} \, I_\pm^\nu$ with $\nu = (1 - \beta)/2$. Substituting into (4):

$$u_t^\star \; = \; \gamma^{-1} \, c_\beta^{-1} \, (D_+^\nu \, \zeta)(t), \qquad \zeta_s \; = \; \big(D_-^\nu \, \bar{\alpha}(s, \cdot)\big)(s), \tag{5}$$

a composition of anticausal and causal Marchaud fractional derivatives of order ν applied to the forecast curve. Without the adapted projection the composition $D_+^\nu \, D_-^\nu$ collapses (in the frequency domain) to the symmetric Riesz derivative of order $2\nu = 1 - \beta$; the projection separates the two half-order pieces and encodes the causality constraint. On a finite horizon the Gohberg–Krein factors introduce terminal-endpoint weight conjugations, giving the boundary-deformed formula of Corollary 2 and convergence to (5) at rate $O(d(t)^{-\nu})$ in the interior (Proposition 3). The empirical range $\beta \in (0.2, 0.6)$ places each half-order ν in $(0.2, 0.4)$.

2 The Gain–Cost Problem in Operator Form

2.1 Setup

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in\mathbb{R}}, \mathbb{P})$ satisfying the usual conditions. The signal α is a mean-zero, progressively measurable process with stationary spectral density $S_\alpha(\xi)$ satisfying $\int(1+|\xi|^{2(1-\beta)+\epsilon})\, S_\alpha(\xi)\, d\xi < \infty$ for some $\epsilon > 0$. Let $L^2 = L^2(\Omega \times \mathbb{R})$ with reference measure $\mathbb{P} \otimes dt$, and let $L^2_{\text{adap}} \subset L^2$ denote the closed subspace of adapted processes: those u with $u_t \in \mathcal{F}_t$ for a.e. t. Denote by P_+ the L^2-orthogonal projection onto L^2_{adap}; pointwise,

$$(P_+X)_s \;=\; \mathbb{E}_s[X_s],$$

which coincides with the optional projection restricted to L^2. Our Fourier convention is $\hat{f}(\xi) = \int e^{i\xi t} f(t)\, dt$; a causal function (support in $t \geq 0$) has Fourier transform analytic in the upper half-plane.

The cost operator C is the deterministic convolution against a symmetric kernel G,

$$(Cu)(t) \;=\; \int G(|t-v|)\, u_v\, dv, \qquad \hat{C}(\xi) \geq 0 \text{ on } \xi \neq 0,$$

acting on L^2 diagonally in ω. Positive-definiteness of $\hat{C}$ ensures strict convexity of the quadratic penalty in (1) on L^2_{adap}. For the pure power-law kernel C is unbounded on $L^2(\mathbb{R})$ and is interpreted between homogeneous Sobolev spaces $C\colon \dot{H}^{-\nu} \to \dot{H}^{\nu}$ with $\nu = (1-\beta)/2$; the technical domain fix is developed in §4. On $[0,T]$ the kernel is weakly singular and G_T is bounded on $L^2([0,T])$ without further regularization.

The *forecast curve* at time t is

$$\bar{\alpha}(t,s) \;=\; \begin{cases} \alpha_s, & s \leq t, \\ \mathbb{E}_t[\alpha_s], & s > t. \end{cases} \tag{6}$$

2.2 The adapted first-order condition

Taking the Gâteaux derivative of the objective in (1) against adapted variations $\delta u \in L^2_{\text{adap}}$ and using $\mathbb{E}[u\, \mathbb{E}_t[\cdot]] = \mathbb{E}[u\cdot]$ for adapted u gives the *adapted first-order condition*

$$\gamma\, \mathbb{E}_t\big[(Cu^\star)(t)\big] \;=\; \alpha_t \quad \text{for a.e. } t. \tag{7}$$

Equivalently in operator form: $\gamma\, P_+CP_+\, u^\star = \alpha$ with $u^\star, \alpha \in L^2_{\text{adap}}$.

The Hessian $\gamma\, P_+CP_+$ is a symmetric strictly positive operator on L^2_{adap} (strict convexity follows from $\hat{C} > 0$ on $\xi \neq 0$ via Plancherel applied to the extension by zero of any adapted u). It admits a bounded two-sided inverse, and (7) has a unique solution

$$u^\star \;=\; \gamma^{-1}\, (P_+\, C\, P_+)^{-1}\, \alpha. \tag{8}$$

The remainder of the paper computes the projected inverse in closed form.

3 Factorization of the Cost Operator

The strategy is to split C as a causal-anticausal product and reduce the projected inverse to a three-step alternating operator. This section states the factorization in both settings — whole line and finite interval — and proves the projected-inverse identity that underlies the closed-form solution.

3.1 Whole line: Wiener–Hopf factorization

Proposition 1 (Wiener–Hopf factorization)**.** *Let $\hat{C}(\xi) \geq 0$ with $\log \hat{C}(\xi)/(1+\xi^2) \in L^1(\mathbb{R})$. Then $\hat{C}$ admits a factorization*

$$\hat{C}(\xi) \ = \ \hat{C}_-(\xi)\, \hat{C}_+(\xi), \tag{9}$$

with $\hat{C}_+$ analytic and non-vanishing in the closed upper half-plane, $\hat{C}_-$ analytic in the closed lower half-plane, and $\overline{\hat{C}_+(\xi)} = \hat{C}_-(\xi)$; equivalently $C_+^* = C_-$. In the time domain, C_+ has kernel supported on $\{s \leq t\}$ (causal Volterra) and C_- on $\{s \geq t\}$ (anticausal). Under the normalization $\overline{\hat{C}_+} = \hat{C}_-$ the factorization is unique up to a unimodular multiplicative constant.

Existence is classical (Wiener and Hopf 1931, Krein 1962). Kernel-specific factors are worked out in §4 (power-law) and §5.1 (exponential).

3.2 Finite interval: Gohberg–Krein factorization

On $[0,T]$ translation invariance is lost, Fourier factorization does not apply, and the appropriate analog is the Volterra-triangular factorization of Gohberg and Krein (1970), abstractly the Arveson outer factorization on a continuous nest (Arveson 1975).

Proposition 2 (Gohberg–Krein factorization)**.** *Let G_T be the restriction of C to $L^2([0,T])$, viewed as an integral operator with weakly singular kernel $G(|t-v|)\mathbf{1}_{[0,T]^2}$, and suppose G_T is strictly positive on $L^2([0,T])$ (injective with $\langle G_T f, f\rangle > 0$ for $f \neq 0$, guaranteed by $\hat{C} > 0$ on $\xi \neq 0$ via Plancherel on the extension by zero). Then G_T admits a causal–anticausal factorization*

$$G_T \ = \ C_- \, C_+, \tag{10}$$

with C_+ a Volterra operator whose kernel is supported on $\{s \leq t\} \cap [0,T]^2$ (causal) and $C_- = C_+^*$ its adjoint (kernel on $\{s \geq t\}$). The causal factor C_+ is unique up to a left unitary commuting with the nest $\{P_{[0,t]}\}$ (a diagonal phase). For the power-law kernel G_T is compact with unbounded inverse, so the factors act between weighted Sobolev spaces rather than on L^2 (§4.1).

The classical Gohberg–Krein outer factor T realizes the Arveson outer factorization of G_T with respect to the continuous nest $\{P_{[0,t]}\}_{t\in[0,T]}$ as $G_T = T\,T^*$ with T causal (Arveson 1975, Thm 4.4.2)(Porter and Stirling 1990, Ex. 6.2/9.2). This places the causal factor on the *left*; the adapted projected inverse of Lemma 1 (below) instead needs the causal factor on the right, obtained by the time reflection $C_+ = RT^*R$, $(Rf)(t) = f(T-t)$, since $RG_TR = G_T$ gives $C_-C_+ = R\,(TT^*)\,R = G_T$. The explicit power-law factors are given in §4.

3.3 The adapted projected inverse

Lemma 1 (Adapted inverse via causal-anticausal factorization)**.** *Let $\mathcal{H}$ be either $L^2(\mathbb{R})$ or $L^2([0,T])$, tensored with $L^2(\Omega)$ against the filtered measure. Let $C = C_-\,C_+$ be a factorization of a symmetric strictly positive operator on $\mathcal{H}$ into a causal Volterra factor C_+ (kernel supported on $\{s \leq t\}$) and its adjoint $C_- = C_+^*$. Let P_+ be the L^2-orthogonal projection onto adapted processes. Then on the adapted subspace,*

$$(P_+\,C\,P_+)^{-1} \ = \ C_+^{-1}\,P_+\,C_-^{-1}. \tag{11}$$

Proof. Let $u \in L^2_{\text{adap}}$ be arbitrary. We verify that $C_+^{-1}P_+C_-^{-1}$ is a two-sided inverse of P_+CP_+ on L^2_{adap}.

Step 1: triangularity relations. Causality of C_+ means its kernel is supported on $\{s \le t\}$, equivalently C_+ is a lower-triangular operator with respect to the nest $\{P_{[-\infty,t]}\}$. This gives the first triangularity relation

$$P_+^\perp\, C_+\, P_+ = 0. \tag{12}$$

Taking adjoints, with $C_- = C_+^*$ and $(P_+^\perp)^* = P_+^\perp$:

$$P_+\, C_-\, P_+^\perp = 0. \tag{13}$$

Step 2: C_+^{-1} preserves L^2_{adap}. For any $f \in L^2_{\mathrm{adap}}$, (12) gives $C_+ P_+^\perp h = 0$ for $h := C_+^{-1} f - P_+ C_+^{-1} f = P_+^\perp C_+^{-1} f$, hence $h = P_+^\perp C_+^{-1} f = 0$ by injectivity of C_+^{-1} (which follows from strict positivity of C and $C_- = C_+^*$). Thus $C_+^{-1} f \in L^2_{\mathrm{adap}}$.

Step 3: $P_+ C_-^{-1}$ maps L^2_{adap} into L^2_{adap}. For $u \in L^2_{\mathrm{adap}}$, write $C_-^{-1} u = P_+ C_-^{-1} u + P_+^\perp C_-^{-1} u$. Apply C_- to both sides: $u = C_- P_+ C_-^{-1} u + C_- P_+^\perp C_-^{-1} u$. Since $u \in L^2_{\mathrm{adap}}$ and $P_+ u = u$, taking P_+ of the equation and using (13) ($P_+ C_- P_+^\perp = 0$) gives $u = P_+ C_- P_+ C_-^{-1} u = P_+ C_-\, P_+ C_-^{-1} u$, so the adapted part $P_+ C_-^{-1} u$ inverts $P_+ C_-$ on L^2_{adap}.

Step 4: right inverse. For $u \in L^2_{\mathrm{adap}}$, using Steps 2–3 and $C = C_- C_+$:

$$(P_+ C P_+)\,(C_+^{-1} P_+ C_-^{-1} u) = P_+\, C_-\, C_+\, C_+^{-1}\, P_+ C_-^{-1} u = P_+\, C_-\, P_+ C_-^{-1} u = u,$$

where the last equality is Step 3.

Step 5: left inverse. For $u \in L^2_{\mathrm{adap}}$:

$$C_+^{-1} P_+ C_-^{-1}\,(P_+ C P_+ u) = C_+^{-1} P_+ C_-^{-1}\, P_+ C_- C_+ u = C_+^{-1} P_+ (P_+ C_-^{-1} P_+)\, C_-\, C_+ u.$$

By Step 3, $P_+ C_-^{-1} P_+$ is the inverse of $P_+ C_-$ on L^2_{adap}; since $C_+ u \in L^2_{\mathrm{adap}}$ (Step 2, applied to u, using the invertible case), $P_+ C_-^{-1} P_+ C_- C_+ u = C_+ u$. Thus the expression equals $C_+^{-1} C_+ u = u$.

Only causality of C_+, strict positivity of C, and $C_- = C_+^*$ enter. The argument covers the whole-line Wiener–Hopf factorization and the finite-interval Gohberg–Krein/Arveson factorization; in the latter case C_+ is the reflected causal Volterra factor of Proposition 2 and $C_- = C_+^*$. Full domain-specific details for the power-law unbounded-operator case are in the appendix. □

3.4 Closed-form optimal trading rate

Combining (8) and (11):

Theorem 1 (Closed-form optimal rate). *Under the hypotheses of §2.1, the unique adapted solution of (7) is*

$$u^\star \;=\; \gamma^{-1}\, C_+^{-1}\, P_+\, C_-^{-1}\, \alpha, \tag{14}$$

understood in either setting: on the whole line with $C_\pm$ the Wiener–Hopf factors, or on $[0,T]$ with $C_+ = RT^*R$ the reflected (terminal-anchored) causal Gohberg–Krein factor of G_T and $C_- = C_+^*$. When linear position constraints are imposed, α is replaced by an effective signal $\alpha^{\mathrm{eff}} = \alpha + \sum_k \mu_k \psi_k$ adjoining one multiplier per constraint: a finite family for expectation constraints such as $\mathbb{E}[X_T] = 0$, and a process-valued multiplier $\mu(\cdot)$ for the pathwise liquidation constraint $X_T = 0$ a.s. (Forde et al. 2022).

The acausal factor C_-^{-1} applied to the adapted signal α at time s requires the future path $\{\alpha_r\}_{r \ge s}$, which is not in $\mathcal{F}_s$. The adapted projection replaces this future by its conditional expectation, and commutation of the deterministic operator C_-^{-1} with conditional expectation on the argument variable (Hytönen et al. 2016, Prop. 2.6.13) gives

$$\big(P_+\, C_-^{-1}\, \alpha\big)_s \;=\; \big(C_-^{-1}\, \bar{\alpha}(s,\cdot)\big)(s). \tag{15}$$

The optimal rate is therefore

$$u_t^\star \;=\; \gamma^{-1}\, C_+^{-1}\, \zeta\,\big|_t, \qquad \zeta_s \;=\; \big(C_-^{-1}\,\bar\alpha(s,\cdot)\big)(s). \tag{16}$$

The forecast curve enters only through the inner C_-^{-1} step; C_+^{-1} then samples the adapted process ζ over $s \le t$, producing an $\mathcal{F}_t$-measurable rate.

4 Power-Law Kernel

For $G(t) = |t|^{-\beta}$ with $\beta \in (0,1)$ and $\nu = (1-\beta)/2$, the Wiener–Hopf and Gohberg–Krein factors are explicit fractional integrals, and Theorem 1 reduces to a fractional derivative of the forecast curve on the whole line and to a boundary-deformed fractional derivative on $[0,T]$.

4.1 Fractional-integral factors

Whole-line factors. The symbol $\hat C(\xi) = c_\beta|\xi|^{\beta-1} = c_\beta(i\xi)^{-\nu}(-i\xi)^{-\nu}$ (standard branches) factors as $\hat C_+(\xi) = c_\beta^{1/2}(-i\xi)^{-\nu}$ (analytic in the upper half-plane) and $\hat C_-(\xi) = c_\beta^{1/2}(i\xi)^{-\nu}$. In the time domain, $C_\pm$ are the causal and anticausal Riemann–Liouville fractional integrals of order ν:

$$C_+ \;=\; c_\beta^{1/2}\, I_+^\nu, \quad (I_+^\nu f)(t) = \tfrac{1}{\Gamma(\nu)}\int_{-\infty}^{t} (t-s)^{\nu-1} f(s)\, ds; \qquad C_- \;=\; c_\beta^{1/2}\, I_-^\nu. \tag{17}$$

Inverses are the Marchaud fractional derivatives $C_\pm^{-1} = c_\beta^{-1/2}\, D_\pm^\nu$ (Samko et al. 1993, §5.4).

Boundedness caveat. The pure power-law symbol satisfies the log-integrability hypothesis of Proposition 1, so the symbol factorization (9) exists distributionally. The factors are however unbounded on $L^2(\mathbb{R})$: $\hat C_\pm(\xi)$ diverges at $\xi = 0$ and $\hat C_\pm^{-1}$ diverges as $|\xi| \to \infty$, so the projected-inverse identity of §3.3 is an equality between unbounded operators. It holds on the dense domain fixed by the standing spectral hypothesis of §2.1, which ensures $C^{-1}\alpha \in L^2$ and gives sense to each Marchaud composition. A bounded-operator setup is recovered by adjoining a temporary-impact term $\frac{1}{2}\eta u_t^2$ (§5.2): the modified symbol $M(\xi) = c_\beta|\xi|^{\beta-1} + \eta/\gamma$ has $M^{-1} \in L^\infty(\mathbb{R})$, Proposition 1 applies with $C_\pm^{-1}$ bounded on L^2 without spectral-decay assumptions on α, and the $\eta \to 0$ limit recovers the pure power-law formulas on the §2.1 domain.

Finite-interval factors. On $[0,T]$ the classical Gohberg–Krein outer construction (Forde et al. 2022, Porter and Stirling 1990) gives $G_T = T\,T^*$ with

$$(T\varphi)(t) \;=\; \int_0^t k(s,t)\,\varphi(s)\, ds, \qquad k(s,t) \;=\; c_\beta^{1/2}\,(t/s)^\nu\,(t-s)^{\nu-1}/\Gamma(\nu), \tag{18}$$

equivalently $T = c_\beta^{1/2}\, B^{-1}\, I_+^\nu\, B$ with $B(t) = t^{-\nu}$, the left-endpoint-anchored form. This outer factorization places the causal factor on the *left*; the adapted projected inverse of Lemma 1 requires the causal factor on the right, $G_T = C_-\, C_+$. By the reflection $(Rf)(t) = f(T-t)$ and $R\, G_T R = G_T$, that factor is the terminal-anchored causal Volterra operator $C_+ = R\,T^* R$,

$$C_+ \;=\; c_\beta^{1/2}\, B_T\, I_+^\nu\, B_T^{-1}, \qquad C_- = C_+^* = c_\beta^{1/2}\, B_T^{-1}\, I_-^\nu\, B_T, \qquad B_T(t) = (T-t)^{-\nu}, \tag{19}$$

with kernel $c_\beta^{1/2}\big((T-s)/(T-t)\big)^\nu (t-s)^{\nu-1}/\Gamma(\nu)$ on $\{s \le t\}$, so that $G_T = C_- C_+ = c_\beta\, B_T^{-1}\, I_-^\nu\, B_T^2\, I_+^\nu\, B_T^{-1}$ (verified by direct kernel integration). Both factors act boundedly between the weighted Sobolev spaces of Porter–Stirling (Porter and Stirling 1990, Ex. 6.2/9.2) and Samko–Kilbas–Marichev (Samko et al. 1993, §13.5). The terminal-anchored weight B_T absorbs the loss of translation invariance at the forward endpoint and pins C_+ to the nest $\{P_{[0,t]}\}$ up to a diagonal phase, breaking manifest $t \leftrightarrow T-t$ symmetry while $G_T = C_- C_+$ preserves it.

4.2 Bulk formula on the whole line

Substituting $C_\pm^{-1} = c_\beta^{-1/2}\, D_\pm^\nu$ into (16):

Corollary 1 (Power-law bulk formula)**.** *For $G(t) = |t|^{-\beta}$ with $\beta \in (0,1)$, $\nu = (1-\beta)/2$, and α stationary adapted satisfying the standing spectral hypothesis of §2.1,*

$$u_t^\star \;=\; \gamma^{-1}\, c_\beta^{-1}\, (D_+^\nu\, \zeta)(t), \qquad \zeta_s \;=\; \big(D_-^\nu\, \bar{\alpha}(s,\cdot)\big)(s). \tag{20}$$

The inner fractional differentiation D_-^ν acts on the forecast curve and cancels the frequency dependence of the impact operator; the outer D_+^ν colors the resulting adapted process causally. Only in the anticipative case (future signal known, no projection) would the whitened process $C_-^{-1}\alpha$ carry the flat-in-impact spectrum $c_\beta^{-1}|\xi|^{1-\beta}\, S_\alpha(\xi)$; the adapted $\zeta_s = (D_-^\nu \bar{\alpha}(s,\cdot))(s)$ replaces future values by forecasts and has a strictly smaller spectrum (for OU, $\zeta_s = \theta^\nu \alpha_s$ with spectrum $\theta^{2\nu} S_\alpha(\xi)$, cf. (21)). Without the projection the composition $D_+^\nu D_-^\nu$ would reduce (in symbol form) to the symmetric Riesz derivative $|\xi|^{1-\beta}$; adaptedness splits it into the two one-sided derivatives of order ν separated by P_+.

For OU α with mean-reversion rate θ, direct Marchaud integration against the exponential forecast tail gives $(D_-^\nu \bar{\alpha}(t,\cdot))(t) = \theta^\nu \alpha_t$, so

$$u_t^{\star,\,\mathrm{OU}} \;=\; \gamma^{-1}\, c_\beta^{-1}\, \theta^\nu\, (D_+^\nu \alpha)(t), \qquad \mathbb{E}\big[u_t^{\star,\,\mathrm{OU}} \;\big|\; \alpha_t\big] \;=\; \gamma^{-1}\, c_\beta^{-1}\, \theta^{1-\beta}\, \alpha_t, \tag{21}$$

positive for every $\beta \in (0,1)$ and every $\theta > 0$: the scale-free causal inverse has no zero on the imaginary axis, so no sign-flip phase transition occurs as θ varies (contrast the exponential kernel, §5.1). The OU signal satisfies the standing spectral hypothesis of §2.1 for $\beta > 1/2$; for $\beta \le 1/2$ the OU spectrum $\sigma^2/(\theta^2+\xi^2)$ fails high-frequency integrability against $|\xi|^{2(1-\beta)+\epsilon}$, and (21) is made rigorous by adding a temporary-impact regularization (§5.2) or by working with increments of α.

4.3 Boundary-deformed formula on a finite interval

Substituting (14) into Theorem 1:

Corollary 2 (Power-law finite-interval formula)**.** *For $G(t) = |t|^{-\beta}$ on $[0,T]$, the adapted optimal rate is*

$$u_t^{\star,T} \;=\; \gamma^{-1}\, c_\beta^{-1}\, \big(B_T\, D_+^\nu\, B_T^{-1}\, P_+\, B_T^{-1}\, D_-^\nu\, B_T\, \alpha^{\mathrm{eff}}\big)(t), \tag{22}$$

where $B_T(t) = (T-t)^{-\nu}$ is the terminal-anchored endpoint weight from the factors (19) *(the multiplications by B_T commute with P_+ since B_T is deterministic, so $B_T^{-1} P_+ B_T^{-1} = P_+ B_T^{-2}$) and $\alpha^{\mathrm{eff}} = \alpha + \sum_k \mu_k \psi_k$ is the effective signal adjoining a multiplier per position constraint — a finite linear system for expectation constraints, and a process-valued multiplier for the pathwise liquidation constraint $X_T = 0$ a.s. (Forde et al. 2022).*

The composition applies, in order: (i) terminal weight B_T to the effective signal; (ii) anticausal Marchaud derivative D_-^ν; (iii) reweighting by B_T^{-1}; (iv) adapted projection; (v) reweighting by B_T^{-1}; (vi) causal Marchaud derivative D_+^ν; (vii) terminal weight B_T. The multiplications arise from substituting the reflected Gohberg–Krein factors $C_\pm$ of (19) into $u^\star = \gamma^{-1} C_+^{-1} P_+ C_-^{-1} \alpha^{\mathrm{eff}}$. Forde, Sánchez-Betancourt and Smith (Forde et al. 2022, Thm 2.2) solve the same finite-interval problem through the full inverse $G_T^{-1} = (T^*)^{-1} T^{-1}$, for which the factor order is immaterial; (22) instead exposes the adapted projection between the two half-order factors, which requires the causal factor on the right.

Formula (22) is a *boundary-deformed fractional derivative*: the whole-line composition $D_+^\nu\, P_+\, D_-^\nu$ of (20) is preserved as the operator core; the terminal weights B_T, B_T^{-1} sandwich it, while the left endpoint enters through the truncation of the Marchaud kernels to $[0,T]$.

4.4 Interior asymptotic and rate of convergence

As $T \to \infty$, (22) converges pointwise to (20) in the interior. Let $d(t) = \min(t, T - t)$ be the distance from t to the nearest boundary. Centering at the midpoint $t = T/2 + t'$, the terminal weight $B_T(T/2+t') = (T/2-t')^{-\nu}$ factors as a T-dependent scalar $(T/2)^{-\nu}$ times a slowly varying interior factor $(1-2t'/T)^{-\nu}$. The scalar cancels in the $B_T^{-1} \cdot B_T$ conjugations of (22), and the interior factor converges to 1 uniformly on compact t' sets, with relative deviation $O(d(t)^{-1})$. The reflected Volterra kernel $\big((T-s)/(T-t)\big)^{\nu}(t-s)^{\nu-1}$ satisfies $\big((T-s)/(T-t)\big)^{\nu} \to 1$ in the interior, converging to the whole-line causal Riemann–Liouville kernel $(t-s)^{\nu-1}/\Gamma(\nu)$.

Proposition 3 (Interior error bound)**.** *For α bounded and effective signal $\alpha^{\mathrm{eff}} \in \dot{H}^{\nu}(\mathbb{R})$,*

$$\left|u_t^{\star,T} - u_t^{\star,\mathbb{R}}\right| \;\le\; C_1(\beta)\,\|\alpha\|_{L^\infty}\, d(t)^{-\nu} \;+\; C_2(\beta)\,\|\alpha^{\mathrm{eff}}\|_{\dot{H}^{\nu}}\, T^{-\nu}\, d(t)^{-\nu}, \tag{23}$$

with $\nu = (1-\beta)/2$ and $C_1(\beta), C_2(\beta)$ β-dependent constants.

The first term is the Marchaud truncation error from cutting the fractional-derivative tail at the boundary; it dominates the weight-conjugation deviation, which is $O(d(t)^{-1})$ and hence subdominant for $\nu < 1$. The second term is the endpoint-constraint correction, entering through the Söhngen–Tricomi eigenfunction $\phi_1(t) = [t(T-t)]^{(\beta-1)/2}$ of the no-signal solution (Gatheral et al. 2012, Forde et al. 2022), whose coefficient is $O(\|\alpha^{\mathrm{eff}}\|_{\dot{H}^{\nu}}\, T^{-\nu})$ under the stated regularity. For empirical $\beta \in (0.2, 0.6)$, $\nu \in (0.2, 0.4)$, giving slow $d(t)^{-\nu}$ interior convergence — a quantitative expression of the long memory of the impact kernel.

5 Discussion

5.1 Exponential kernel and the signal-speed response

The exponential kernel $G(t) = e^{-\kappa|t|}$ is a tractable alternative to the power-law. It bounds the cost operator on $L^2(\mathbb{R})$ and underlies the Obizhaeva–Wang (2013) resilience model and much of the Almgren–Chriss line of work. Working through it in the present framework does two things: it removes the domain subtleties that arise for the power-law kernel in §4, and it makes visible which features of Theorem 1 are structural (the three-step factorization) and which are specific to the fractional case (unbounded factors, fractional-derivative discretization, absence of a sign-flip regime).

The symbol

$$\hat{C}(\xi) \;=\; \frac{2\kappa}{\kappa^2+\xi^2} \;=\; \frac{\sqrt{2\kappa}}{\kappa - i\xi}\cdot\frac{\sqrt{2\kappa}}{\kappa+i\xi}$$

splits into rational one-sided factors that are bounded on $L^2(\mathbb{R})$. Their inverses are first-order differential operators,

$$C_+^{-1} \;=\; (2\kappa)^{-1/2}(\kappa+\partial_t), \qquad C_-^{-1} \;=\; (2\kappa)^{-1/2}(\kappa-\partial_t),$$

and substituting into Theorem 1 gives the bulk formula

$$u_t^{\star,\exp} \;=\; \frac{1}{2\kappa\gamma}\,(\kappa+\partial_t)\,\zeta_t, \qquad \zeta_s \;=\; (\kappa-\partial_r)\,\bar{\alpha}(s,r)\big|_{r=s^+}. \tag{24}$$

The optimal rate is a local first-order combination of the forecast curve and its time derivative — a striking simplification of the general nonlocal recipe, made possible by the rational symbol.

For an Ornstein–Uhlenbeck signal with mean reversion θ, the whitened process reduces to $\zeta_s = (\kappa+\theta)\alpha_s$, and (24) yields

$$u_t^{\star,\exp} \;=\; \frac{\kappa+\theta}{2\kappa\gamma}\big[(\kappa-\theta)\alpha_t + \sigma\dot W_t\big], \qquad \mathbb{E}\big[u_t^{\star,\exp} \;\big|\; \alpha_t\big] \;=\; \frac{\kappa^2-\theta^2}{2\kappa\gamma}\,\alpha_t. \tag{25}$$

The white-noise term $\sigma\dot W_t$ reflects that the exponential-kernel cost does not penalize the high-frequency content of the rate, so $u^{\star,\exp}$ is a distribution-valued optimum; it is regularized to an L^2 process by any temporary-impact term $\frac{1}{2}\eta u_t^2$ (§5.2), the empirically relevant case. The conditional mean rate flips sign at $\theta=\kappa$. When the signal mean-reverts faster than the impact tail decays ($\theta>\kappa$), the trader takes the opposite side of the signal on average: any signal-following trade would leave an impact residual longer-lived than the signal itself, so acting against the signal amortises the cost. The power-law analog (21) shows no such transition; the scale-free causal inverse has a branch point at $\xi=0$ and no zero on the imaginary axis, so no pole–zero crossing occurs as θ varies.

Both statements are instances of a kernel-general formula. Let α be a stationary OU signal with rate θ generating its own filtration, and define the *signal-speed response* $R(\theta) := \lim_{q\downarrow 0}\mathbb{E}[u^\star_{t+q} \mid \alpha_t]/\alpha_t$; the innovations arriving in $(t,t+q]$ are independent of α_t, so $R(\theta)$ is the average trade direction executed next by a trader holding the current signal value. Write

$$\Phi(\theta) \;:=\; \hat C_+(i\theta) \;=\; \exp\Big[\frac{\theta}{2\pi}\int_{-\infty}^{\infty}\frac{\log\hat C(t)}{\theta^2+t^2}\,dt\Big] \;>\; 0, \tag{26}$$

the causal factor continued to the Laplace point of the signal; the Szegő/outer representation shows Φ is real and positive because $\hat C$ is even and positive, and $\hat C_-(-i\theta)=\Phi(\theta)$ by conjugate symmetry. In the normalization of Theorem 1 the whitened process is $\zeta_s = \alpha_s/\Phi(\theta)$, consistent with $\zeta = c_\beta^{-1/2}\theta^\nu\alpha$ for the power law and, after the rescaling absorbed into (24), with $\zeta=(\kappa+\theta)\alpha$. For symbols that are rational (finite mixtures of exponentials) or power-law,

$$R(\theta) \;=\; \frac{1}{\gamma\,\Phi(\theta)}\Big[\frac{1}{\Phi(\theta)} - 2c_1\theta\Big], \qquad c_1 \;=\; \lim_{|\xi|\to\infty}\big[(-i\xi)\,\hat C_+(\xi)\big]^{-1}, \tag{27}$$

derived in the appendix. The constant c_1 measures the derivative loading of the inverse causal factor: $c_1 = (-2G'(0^+))^{-1/2}$ for kernels with a kink at the origin ($\hat C(\xi)\sim -2G'(0^+)\,\xi^{-2}$ at high frequency), and $c_1=0$ for the power-law cusp. The exponential kernel has $\Phi=\sqrt{2\kappa}/(\kappa+\theta)$ and $c_1=(2\kappa)^{-1/2}$, and (27) returns $(\kappa^2-\theta^2)/2\kappa\gamma$; the power-law kernel has $\Phi(\theta)^2 = c_\beta\theta^{\beta-1} = \hat C(\theta)$, and (27) returns $\theta^{1-\beta}/\gamma c_\beta$ — the cost symbol inverted at real frequency equal to the signal speed.

Formula (27) settles when contrarian trading is optimal: $\operatorname{sign} R(\theta) = \operatorname{sign}\big(1-2c_1\theta\,\Phi(\theta)\big)$. For kink kernels $2c_1\theta\,\Phi(\theta)\to 2$ as $\theta\to\infty$, so every sufficiently fast signal is traded against; for the power-law cusp $R(\theta)>0$ at every signal speed. The two-exponential mixture $G(t) = w_1e^{-\kappa_1|t|} + w_2e^{-\kappa_2|t|}$ has $\hat C_+(\xi) = \sqrt{2W}\,(\mu-i\xi)/[(\kappa_1-i\xi)(\kappa_2-i\xi)]$ with $W=w_1\kappa_1+w_2\kappa_2$ and $\mu^2 = (w_1\kappa_1\kappa_2^2 + w_2\kappa_2\kappa_1^2)/W$, giving

$$R(\theta) \;=\; \frac{(\kappa_1+\theta)(\kappa_2+\theta)\,\big[(\kappa_1+\theta)(\kappa_2+\theta) - 2\theta(\mu+\theta)\big]}{2\gamma\,W\,(\mu+\theta)^2}, \tag{28}$$

with exactly one sign change, at $\theta^* = \frac{1}{2}\big[(\kappa_1+\kappa_2-2\mu) + \sqrt{(\kappa_1+\kappa_2-2\mu)^2+4\kappa_1\kappa_2}\big]$: the bracket in (28) is a downward parabola in θ, positive at $\theta=0$. Setting $\kappa_1=\kappa_2=\kappa$ recovers $\theta^*=\kappa$. Discretized adapted optima reproduce (27)–(28) to discretization accuracy. The regression of $u_t^\star$ on the contemporaneous α_t instead equals $1/\gamma\Phi(\theta)^2>0$ for every kernel: the contrarian term $-2c_1\theta$ is carried by the innovation atom in the optimal rate — the white-noise component of (25) — which is present exactly when singular trading rates have finite cost, i.e. when G is bounded at the origin.

5.2 Temporary impact

Adding a temporary-impact term $\frac{1}{2}\eta u_t^2$ modifies the FOC symbol to $M(\xi) = c_\beta|\xi|^{\beta-1} + \eta/\gamma$. The added constant provides high-frequency coercivity that the pure power-law symbol lacks, so $u^\star \in L^2$ without spectral-decay assumptions on α. Wiener–Hopf factorization of M gives modified one-sided factors; the crossover frequency $\xi_* = (\gamma c_\beta/\eta)^{1/(1-\beta)}$ separates a long-memory fractional regime ($|\xi| \ll \xi_*$) from a myopic signal-following regime ($|\xi| \gg \xi_*$) in which $u^\star \approx \alpha/\eta$. The $\eta \to 0$ limit is singular but recovers (20) under the spectral-decay hypothesis of §2.1.

5.3 Multi-asset extension

Empirical cross-impact between correlated assets — trades in one name moving others in the same principal-component sector — motivates a portfolio version of Theorem 1. For a cross-impact kernel $\mathbf{K}(t) = |t|^{-\beta}\mathbf{A}$ with $\mathbf{A} = Q\Lambda Q^\top$ symmetric positive-definite, Theorem 1 diagonalizes in the eigenbasis of $\mathbf{A}$: the scalar fractional-derivative rule (20) applies independently to each principal-component alpha with eigenvalue prefactor Λ_{ii}^{-1}. The symmetric case covers the leading positive-definite parameterization; non-symmetric cross-impact requires a matrix Wiener–Hopf factorization of $\mathbf{A}(\xi)$, which exists under the same log-integrability hypothesis but may carry non-zero partial indices, in which case the factor is not a simple fractional integral and the scalar reduction fails (Gohberg and Krein 1970).

5.4 Numerical implementation

Fractional derivatives discretize to Toeplitz matrices. On a uniform grid of N points, $D_\pm^\nu$ is a lower- or upper-triangular Toeplitz whose entries are generalized binomial coefficients from the expansion of $(1-z)^\nu$. Evaluating (20) over the whole grid costs $O(N \log N)$ via FFT, compared with $O(N^2)$ Nyström inversion of the Fredholm equation. The finite-interval formula (22) adds diagonal multiplications by B_T and B_T^{-1}, preserving the $O(N \log N)$ cost.

5.5 Role of the forecast curve: innovations filter and value

Theorem 1 makes explicit that the optimal rate at time t depends on the *entire* forecast curve $\bar{\alpha}(t,\cdot)$. When α is a stationary, purely non-deterministic Gaussian process generating its own filtration, the dependence collapses to a single transfer function. Let $S_\alpha = |\varphi_+|^2$ with φ_+ the outer spectral factor (Szegő condition $\int \log S_\alpha(\xi)/(1+\xi^2)\,d\xi > -\infty$), so that $\alpha = \varphi_+(D)\dot{W}$ is a causal moving average of its innovations (Wiener 1949), and let Π_+ denote the Riesz projection: truncation of a moving-average kernel to non-negative lags. Conditional expectation of a stationary linear functional of W truncates its kernel at lag zero, so the adapted projection of Theorem 1 becomes Π_+ and

$$u^\star \;=\; \hat{g}(D)\,\dot{W}, \qquad \hat{g} \;=\; \gamma^{-1}\,\hat{C}_+^{-1}\,\Pi_+\big[\hat{C}_-^{-1}\varphi_+\big], \tag{29}$$

with $\hat{g} \in L^2$ under the standing spectral hypothesis (appendix). The optimal rate is a stationary causal filter of the signal innovations — the Wiener–Kolmogorov form of the three-step recipe. Its spectrum is $S_{u^\star} = |\hat{g}|^2$; for the power-law kernel $S_{u^\star}$ vanishes like $|\xi|^{1-\beta}$ at low frequency whenever $S_\alpha(0) < \infty$, so cost-optimal flow is anti-persistent at long horizons.

Value of a forecast. Stationarity makes the value *rate* (expected P&L per unit time) the relevant quantity, $v(u) = \mathbb{E}[u_t\alpha_t] - \frac{\gamma}{2}\mathbb{E}[u_t(Cu)_t]$; at the optimum $v = \frac{1}{2}\mathbb{E}[u_t^\star\alpha_t]$, and (29) evaluates it:

$$v_{\mathrm{ad}} \;=\; \frac{1}{4\pi\gamma}\,\big\|\Pi_+\big(\hat{C}_-^{-1}\varphi_+\big)\big\|_{L^2}^2, \qquad v_{\mathrm{ant}} - v_{\mathrm{ad}} \;=\; \frac{1}{4\pi\gamma}\,\big\|\Pi_-\big(\hat{C}_-^{-1}\varphi_+\big)\big\|_{L^2}^2, \tag{30}$$

where $v_{\text{ant}} = \frac{1}{4\pi\gamma}\int S_\alpha(\xi)/\hat{C}(\xi)\,d\xi$ is the value rate of the perfectly-anticipating optimum $\gamma^{-1}C^{-1}\alpha$ and $\Pi_- = I - \Pi_+$. The half-whitened spectral factor $\hat{C}_-^{-1}\varphi_+$ splits into a causal part, which prices the adapted problem, and an anticausal remainder, whose energy is the exact cost of causality. For the power-law kernel and an OU signal, $\Pi_+((i\xi)^\nu\varphi_+) = \theta^\nu\varphi_+$ (pole cancellation, appendix), giving

$$v_{\text{ad}} \;=\; \frac{\sigma^2\,\theta^{-\beta}}{4\gamma c_\beta}, \qquad \frac{v_{\text{ad}}}{v_{\text{ant}}} \;=\; \sin(\pi\beta/2):$$

adaptedness costs nothing in the short-memory limit $\beta \to 1$ and destroys all value in the permanent-impact limit $\beta \to 0$. The weighting of the signal spectrum by $|\xi|^{1-\beta}$ in v_{ant}, and by the causal-energy functional in v_{ad}, differs from an R^2 of the signal against future returns: two signals with identical R^2 can differ arbitrarily in tradeable value once their timescales differ, and (30) quantifies what makes one forecast better than another for a given impact kernel.

Forecast horizon. The Marchaud representation of D_-^ν averages forecast increments $\bar{\alpha}(s,r) - \bar{\alpha}(s,s)$ over all future lags $r > s$ with weight $(r-s)^{-1-\nu}$. Forecasts are required over all horizons; horizon truncation gives an error decaying like $r_{\max}^{-\nu}$ in the maximum forecast horizon.

5.6 Comparison with the Volterra-equation approach

Abi Jaber and Neuman (2025) characterize the adapted optimum of (1) on $[0,T]$ as the unique solution of the linear stochastic Volterra equation of the second kind

$$\gamma\,u_t^\star + \gamma\int_0^t R(t,s)\,u_s^\star\,ds \;=\; \alpha_t^\star, \tag{31}$$

where $R(t,s)$ is the resolvent kernel of the second kind of the impact propagator G on $[0,T]$ and $\alpha^\star$ is an effective signal incorporating the terminal-inventory multiplier. Existence and uniqueness follow from the Volterra theory and hold for general adapted signals and general Volterra propagators. Closed forms require kernel-specific resolution of the resolvent R.

The present paper gives a complementary characterization. Equation (31) characterizes $u^\star$ through a Neumann-series expansion of the resolvent; the Wiener–Hopf identity (11) instead expresses $u^\star$ directly as the three-step product $\gamma^{-1}C_+^{-1}P_+C_-^{-1}\alpha$, with no resolvent series. The two expressions are equivalent on $[0,T]$ once the reflected Gohberg–Krein factors (19) are substituted: the resolvent expansion in (31) is the Neumann expansion of the inverse Volterra operator, which the factorization bypasses by construction. On the whole line the resolvent-of-the-second-kind characterization is replaced entirely by the Wiener–Hopf bulk formula (20), valid for any stationary adapted signal under the spectral hypothesis of §2.1.

The two approaches differ in what they emphasize. The Volterra-equation approach handles the general finite-horizon problem, terminal constraints, and non-stationary signals; it does not require translation invariance or a stationary assumption. The spectral factorization approach requires the kernel to have a Fourier symbol admitting the Wiener–Hopf decomposition (9), and the bulk formula (20) is valid only in the stationary (whole-line) setting; in exchange, the closed form expresses the operator structure of the adapted inverse explicitly as a composition of one-sided fractional operators, making the fractional-calculus content visible and the order $\nu = (1-\beta)/2$ a direct consequence of the kernel symbol.

For the power-law kernel, the FSS factorization (Forde et al. 2022) gives the finite-interval resolvent $R(t,s)$ as an integral against the kernel of T^{-1} (the inverse left-causal Volterra factor). Corollary 2 gives instead $u^{\star,T} = \gamma^{-1}C_+^{-1}P_+C_-^{-1}\alpha^{\text{eff}}$ with the reflected factor $C_+ = RT^*R$; these are equivalent formulations of the same operator, and the FSS formula for R can be recovered from the kernel of $(C_+^{-1}C_-^{-1})$. The Proposition 3 interior bound (23) then expresses how the weight-conjugation error (the difference between C_+ and the whole-line $c_\beta^{1/2}I_+^\nu$) decays as the boundary recedes, quantifying the range of validity of the bulk formula without resolving the full resolvent.

5.7 Joint gain–risk–cost formulation

The joint gain–risk–cost problem inherits the same operator structure once the two frictions share a coordinate. Cost acts on the trading rate u; a mean–variance holding penalty $\frac{\lambda}{2}\mathbb{E}\int x_t^\top \Sigma x_t\,dt$ acts on the position $x_t = \int_{-\infty}^t u_s\,ds$, which in rate coordinates is the operator with Fourier symbol $\lambda\Sigma/\xi^2$. Adding it to (1) replaces C by the symbol $\gamma\hat{C}(\xi) + \lambda\Sigma/\xi^2$ — still a positive Toeplitz operator, admitting Wiener–Hopf and Gohberg–Krein factorization, to which Lemma 1 applies without modification. The pure trading limit ($\lambda \to 0$) and the pure holding limit ($\gamma \to 0$) are the two boundaries of this joint factorization.

6 Concluding Remarks

This paper establishes a closed-form solution to the gain–cost trading problem under a general symmetric positive-definite propagator kernel. The construction rests on the causal–anticausal factorization of the cost operator — Wiener–Hopf on the whole line, Gohberg–Krein on a finite interval, both realized as instances of the Arveson outer factorization for positive operators on a continuously nested Hilbert space. The adapted optimal rate is expressed as the composition of the causal square-root inverse of the cost operator with the adapted projection of the anticausal square-root inverse applied to the forecast curve, an expression whose form depends on the kernel only through its factorization.

For the power-law kernel, the whole-line factors are the Liouville (Weyl) fractional integrals of order $\nu = (1-\beta)/2$, and the closed form expresses the optimal rate as a composition of an anticausal and a causal Marchaud fractional derivative of order ν, separated by the adapted projection P_+. Without P_+ the composition would reduce to the symmetric Riesz derivative of order $1-\beta$; the projection preserves the two half-order pieces and encodes the causality constraint. On a finite horizon the fractional integrals acquire terminal-anchored endpoint-weight conjugations induced by the reflected Gohberg–Krein factorization, and the optimal rate takes the form of a boundary-deformed fractional derivative that converges to the whole-line formula in the interior at rate $O(d(t)^{-\nu})$.

A Appendix

Proof of Lemma 1 (whole-line case). With $C_\pm = c_\beta^{1/2}I_\pm^\nu$ on $L^2(\mathbb{R})$, $C_\pm$ are Hilbert-space isomorphisms between $\dot{H}^{-\nu}$ and L^2 (respectively L^2 and $\dot{H}^\nu$) by Plancherel and the symbol $c_\beta^{1/2}|\xi|^{(\beta-1)/2}$. Their adjointness $C_+^* = C_-$ follows from the kernel-flip $(t-s)^{\nu-1}\mathbf{1}_{s\le t} \mapsto (s-t)^{\nu-1}\mathbf{1}_{t\le s}$. On $L^2_{\text{adap}} \cap \dot{H}^{-\nu}$ the composition P_+CP_+ is bounded, symmetric, and strictly positive (from $\hat{C} > 0$ on $\xi \neq 0$), hence boundedly invertible on its range. Causality of C_+ implies $P_+^\perp C_+ P_+ = 0$, so C_+ preserves L^2_{adap} and C_+^{-1} commutes with P_+ on adapted inputs. The adjoint identity $P_+C_-P_+^\perp = 0$ follows. Taking adjoints of $C_+^{-1}\colon L^2 \to \dot{H}^{-\nu}$ gives $P_+C_-^{-1}P_+^\perp = 0$ as well, so C_-^{-1} preserves ${L^2_{\text{adap}}}^\perp$. For adapted u: $P_+CP_+u = P_+C_-C_+u$; applying $C_+^{-1}P_+C_-^{-1}$ returns u via $P_+C_-^{-1}P_+C_-$ acting as P_+ on the adapted vector C_+u.

Proof of Lemma 1 (finite-interval case). Take $C_+ = RT^*R$ the reflected causal Gohberg–Krein factor of Proposition 2 and $C_- = C_+^* = RTR$, so that $C_-C_+ = R(TT^*)R = RG_TR = G_T$. Volterra causality of C_+ (kernel on $\{s \le t\} \cap [0,T]^2$) gives $P_+^\perp C_+ P_+ = 0$ and $P_+ C_- P_+^\perp = 0$; the composition argument of the whole-line case then applies verbatim.

Existence of the outer factor. The Wiener–Hopf factorization on $\mathbb{R}$ under the log-integrability hypothesis is classical (Wiener and Hopf 1931, Krein 1962). The finite-interval outer factorization $G_T = TT^*$ is Arveson's theorem for positive operators on a continuous nest (Arveson 1975, Thm 4.4.2); an equivalent constructive proof for symmetric weakly-singular integral kernels via spectral square root is in (Porter and Stirling 1990, Ex. 6.2/9.2); the explicit left-anchored factor (18) for the power-law kernel is derived in (Forde et al. 2022, pp. 590–591), and the reflected causal factor (19) required here follows from it by the reflection R.

Proof of Corollary 1. Define the candidate $u_t^{\mathrm{cand}} := \gamma^{-1}c_\beta^{-1}(D_+^\nu\zeta)(t)$. Adaptedness: D_+^ν at time t depends only on $\{\zeta_s\}_{s\le t}$, and each $\zeta_s \in \mathcal{F}_s \subset \mathcal{F}_t$ by (15) and $\mathcal{F}_s$-measurability of the forecast curve. FOC verification proceeds in three steps.

Step (a). Conditional Fubini (Klenke 2014, Thm 14.16) applied to the Marchaud representation gives $\mathbb{E}_t[(D_+^\nu\zeta)(v)] = (D_+^\nu\hat{\zeta}_t)(v)$ with $\hat{\zeta}_t(s) := \mathbb{E}_t[\zeta_s]$.

Step (b). For $s \le t$: $\zeta_s \in \mathcal{F}_s \subset \mathcal{F}_t$, so $\hat{\zeta}_t(s) = \zeta_s$. For $s > t$: by tower and conditional Fubini applied at conditioning time t, $\hat{\zeta}_t(s) = (D_-^\nu\bar{\alpha}(t,\cdot))(s)$.

Step (c). The symbol identity $\hat{C}(\xi)(-i\xi)^\nu = c_\beta(i\xi)^{-\nu}$ yields the operator identity $C\,D_+^\nu = c_\beta I_-^\nu$ on $L^2(\mathbb{R})$. Substituting:

$$\gamma\,\mathbb{E}_t\big[(Cu^{\mathrm{cand}})(t)\big] = (I_-^\nu\hat{\zeta}_t)(t) = (I_-^\nu D_-^\nu\bar{\alpha}(t,\cdot))(t) = \bar{\alpha}(t,t) = \alpha_t,$$

using $I_-^\nu D_-^\nu = \mathrm{id}$ on $H^\nu(\mathbb{R})$ (Samko et al. 1993, §5.3 Thm 5.3), applied to the future-half-line piece of the forecast curve, which is in H^ν pathwise under the standing spectral hypothesis, and the fact that I_-^ν at t samples only $\hat{\zeta}_t(s)$ for $s \ge t$, where Step (b) gives the closed form.

Uniqueness. $\hat{C}(\xi) = c_\beta|\xi|^{\beta-1} > 0$ on $\xi \ne 0$, so the quadratic penalty in (1) is strictly convex on L^2_{adap} and the adapted FOC has a unique solution.

Admissibility. The spectral hypothesis ensures $|\xi|^{1-\beta}\widehat{\alpha}(\xi) \in L^2$ pathwise, so $u^\star \in L^2_{\mathrm{adap}}$ by Plancherel. □

Proof of Corollary 2. The anticausal Marchaud derivative D_-^ν in (22) acts on α^{eff} extended by zero outside $[0,T]$, the natural convention for the finite-interval Volterra calculus (Samko et al. 1993, §13.5). Substitute the reflected factors (19) into (14): $u^{\star,T} = \gamma^{-1}\,C_+^{-1}\,P_+\,C_-^{-1}\,\alpha^{\mathrm{eff}}$ with $C_+^{-1} = c_\beta^{-1/2}\,B_T D_+^\nu B_T^{-1}$ and $C_-^{-1} = c_\beta^{-1/2}\,B_T^{-1} D_-^\nu B_T$ (inverting (19) using $(I_\pm^\nu)^{-1} = D_\pm^\nu$). Collecting the two $c_\beta^{-1/2}$ factors into c_β^{-1} gives (22). C_+^{-1} is a Volterra operator between the appropriate weighted Sobolev spaces (Samko et al. 1993, §13.5), not on L^2 directly. Adaptedness follows from causality of C_+^{-1} (Volterra inverse of Volterra is Volterra). □

Proof of Proposition 3. The Marchaud tail bound: $|(D_-^\nu f)(s) - (D_-^{\nu,[0,T]}f)(s)| \le \frac{2\|f\|_\infty}{\Gamma(1-\nu)}(T-s)^{-\nu}$ from

$$(D_-^\nu f)(s) = \tfrac{\nu}{\Gamma(1-\nu)}\int_0^\infty \tfrac{f(s)-f(s+r)}{r^{1+\nu}}\,dr,$$

truncated at $r = T - s$. Symmetrically the outer D_+^ν contributes a tail bounded by $t^{-\nu}$. The dominant interior truncation is on the closer boundary, giving the first term of (23). The terminal-weight conjugation contributes a further error: on the Marchaud support the ratio $B_T(t)/B_T(s) = \big((T-s)/(T-t)\big)^\nu = 1+O((t-s)/(T-t))$, so its deviation from the whole-line unweighted operator is $O(d(t)^{-1})$, subdominant to $d(t)^{-\nu}$ for $\nu < 1$. The endpoint-constraint correction enters through the Söhngen–Tricomi eigenfunction $\phi_1(t) = [t(T-t)]^{(\beta-1)/2}$ of the no-signal solution (Gatheral et al. 2012, Forde et al. 2022), which satisfies $|\phi_1(t)| \le d(t)^{-\nu}(T/2)^{-\nu}$; under $\alpha^{\text{eff}} \in \dot{H}^\nu$ its coefficient is $O(\|\alpha^{\text{eff}}\|_{\dot{H}^\nu}\, T^{-\nu})$, giving the second term of (23). □

Derivation of the innovations form and value (29)–(30). $C_-^{-1}\alpha$ is the stationary filter of W with symbol $h := \hat{C}_-^{-1}\varphi_+$. For fixed s, $\mathbb{E}_s$ annihilates the innovations after s, so $(P_+C_-^{-1}\alpha)_s$ is the filter with kernel truncated to non-negative lags, symbol Π_+h; composing with C_+^{-1} gives (29). Membership: $|\xi|^\nu h = c_\beta^{-1/2}|\xi|^{2\nu}\varphi_+ \in L^2$ by the standing hypothesis, so $h \in H^\nu(\mathbb{R})$; the half-line indicator is a bounded pointwise multiplier on H^ν for $\nu < 1/2$ (Lions and Magenes 1972, Ch. 1), so $\Pi_+h \in H^\nu$ and $\hat{g} = \gamma^{-1}\hat{C}_+^{-1}\Pi_+h \in L^2$. Value: the FOC (7) and adaptedness of $u^\star$ give $\mathbb{E}[u_t^\star\alpha_t] = \gamma\,\mathbb{E}[u_t^\star(Cu^\star)_t]$, hence $v = \frac{1}{2}\mathbb{E}[u_t^\star\alpha_t]$. By the Itô isometry and Plancherel, $\mathbb{E}[u_t^\star\alpha_t] = \frac{1}{2\pi}\int \hat{g}\,\overline{\varphi_+}\,d\xi$; since $\overline{\varphi_+} = \overline{\hat{C}_-\,h} = \hat{C}_+\overline{h}$ on the real axis, the integrand is $\gamma^{-1}(\Pi_+h)\,\overline{h}$, and self-adjointness of Π_+ gives $\gamma^{-1}\|\Pi_+h\|^2/2\pi$, the first formula in (30). The anticipative optimum replaces Π_+h by h; the gap is $\|\Pi_-h\|^2$ by Pythagoras. OU pole cancellation: with $\varphi_+ = \sigma/(\theta - i\xi)$,

$$h - \frac{\varphi_+}{\hat{C}_-(-i\theta)} \;=\; \Big[\frac{1}{\hat{C}_-(\xi)} - \frac{1}{\hat{C}_-(-i\theta)}\Big]\frac{\sigma}{\theta - i\xi}$$

is square-integrable and analytic in the lower half-plane (the pole at $\xi = -i\theta$ is removed and $1/\hat{C}_-$ is analytic there), hence anticausal; so $\Pi_+h = \varphi_+/\hat{C}_-(-i\theta) = \varphi_+/\Phi(\theta)$. For the power-law kernel $\Phi(\theta) = c_\beta^{1/2}\theta^{-\nu}$, and $\|\varphi_+\|^2 = \pi\sigma^2/\theta$, $\|h\|^2 = c_\beta^{-1}\sigma^2\theta^{2\nu-1}\,\pi/\cos(\pi\nu)$ give the displayed v_{ad} and the ratio $\cos(\pi\nu) = \sin(\pi\beta/2)$.

Derivation of the response formula (27). By the pole cancellation above, $\zeta = \alpha/\Phi(\theta)$ and $\hat{g}(\xi) = \gamma^{-1}\sigma/[\Phi(\theta)\,\hat{C}_+(\xi)\,(\theta - i\xi)]$. For $q > 0$, $\text{Cov}(u_{t+q}^\star, \alpha_t) = \sigma\int_0^\infty g(q+r)\,e^{-\theta r}\,dr$ with g the time-domain kernel of $\hat{g}$. For rational symbols $\hat{g}(\xi) \to a_0 := \gamma^{-1}\sigma c_1/\Phi(\theta)$ as $|\xi| \to \infty$, so $g = a_0\delta + g_{\text{reg}}$ with g_{reg} integrable against $e^{-\theta\cdot}$; the atom at lag 0 leaves the integral for every $q > 0$, and dominated convergence gives $\text{Cov} \to \sigma[(\mathcal{L}g)(\theta) - a_0]$ as $q \downarrow 0$, where $(\mathcal{L}g)(\theta) = \hat{g}(i\theta) = \gamma^{-1}\sigma/[2\theta\,\Phi(\theta)^2]$. Dividing by $\text{Var}(\alpha_t) = \sigma^2/2\theta$ yields (27). For the power-law class $\hat{g}(\xi) = O(|\xi|^{\nu-1}) \to 0$, so $a_0 = c_1 = 0$ and $R = 1/\gamma\Phi(\theta)^2$. The same limit computes the position-increment response $\lim_{\Delta\downarrow 0}\mathbb{E}[x_{t+\Delta} - x_t \mid \alpha_t]/\Delta\alpha_t$. □

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